RoboCloud

"The AWS for real robotics"

Speaker name (pronouns)

Speaker job title
Speaker company

Speaker name (pronouns)

Speaker job title
Speaker company

The Physical AI Revolution

4.3M+

Industrial robots in factories worldwide.
The Next Wave: America's Reindustrialization With Physical AI.

2M+

Manufacturing jobs unfilled in US by 2030, due to physically demanding, repetitive, or hard-to-staff tasks.

90%

Self-healing Physical AI systems to reduce downtime by up to 90% by 2030.

$5T

Market for Humanoids by 2050, with 1B+ humanoids launched globally.

“The ChatGPT moment of robot development has arrived. There has been a breakthrough in the field of 'Physical AI'. These models have the ability to understand the real world, reason, and plan actions.”

- Jessen Huang, CEO of Nvidia

Robotics Is Stuck Before Scaled Deployment

$30-150k
unit cost

Humanoid robots remain prohibitively expensive, especially with SDK access. Most teams cannot afford multiple units, limiting experimentation and iteration speed.

4+ weeks
setup time

Deploying a humanoid robot requires significant engineering effort: hardware calibration, environment setup, and software integration. Robots are not plug-and-play—they behave more like complex lab systems than computers.

0-2 robots
for most labs

Access is extremely limited. Robots are shared across teams, leading to scheduling conflicts and delayed experimentation. There is no on-demand access—only queues.

<18%
utilization in R&D

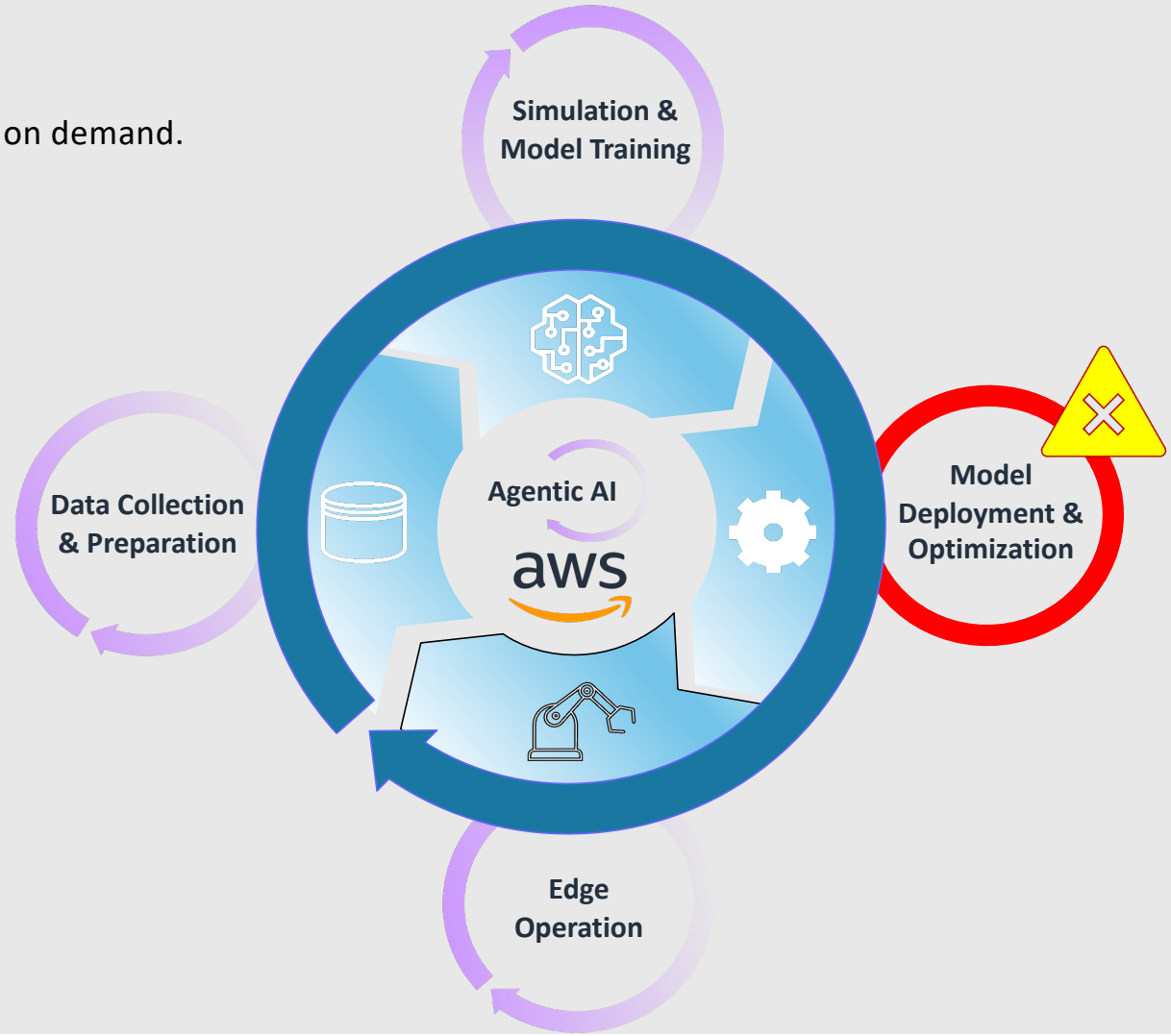
Most robots sit idle outside of short experiment windows.
Time is lost to setup, debugging, and coordination. Expensive hardware remains massively underutilized.

Sim-to-Real is bottleneck

The missing infrastructure is real-world access on demand.

Challenges

- No perfect simulation for real physical world
- Sim-to-Real requires massive amount of tests
- Real world tests are time consuming
- No parallel run due to limited devices/spaces



Solution: On-demand Robot-as-Service

Like AWS—but for physical robots.

HOW IT WORKS

1. **Select a Robot**
   Browse a fleet of multi-brand humanoid robots — pick the platform that fits your experiment.

2. **Choose a Studio**
   Select a controlled testing environment with stairs, objects, adjustable lighting, and more. It allows to run different use-cases.

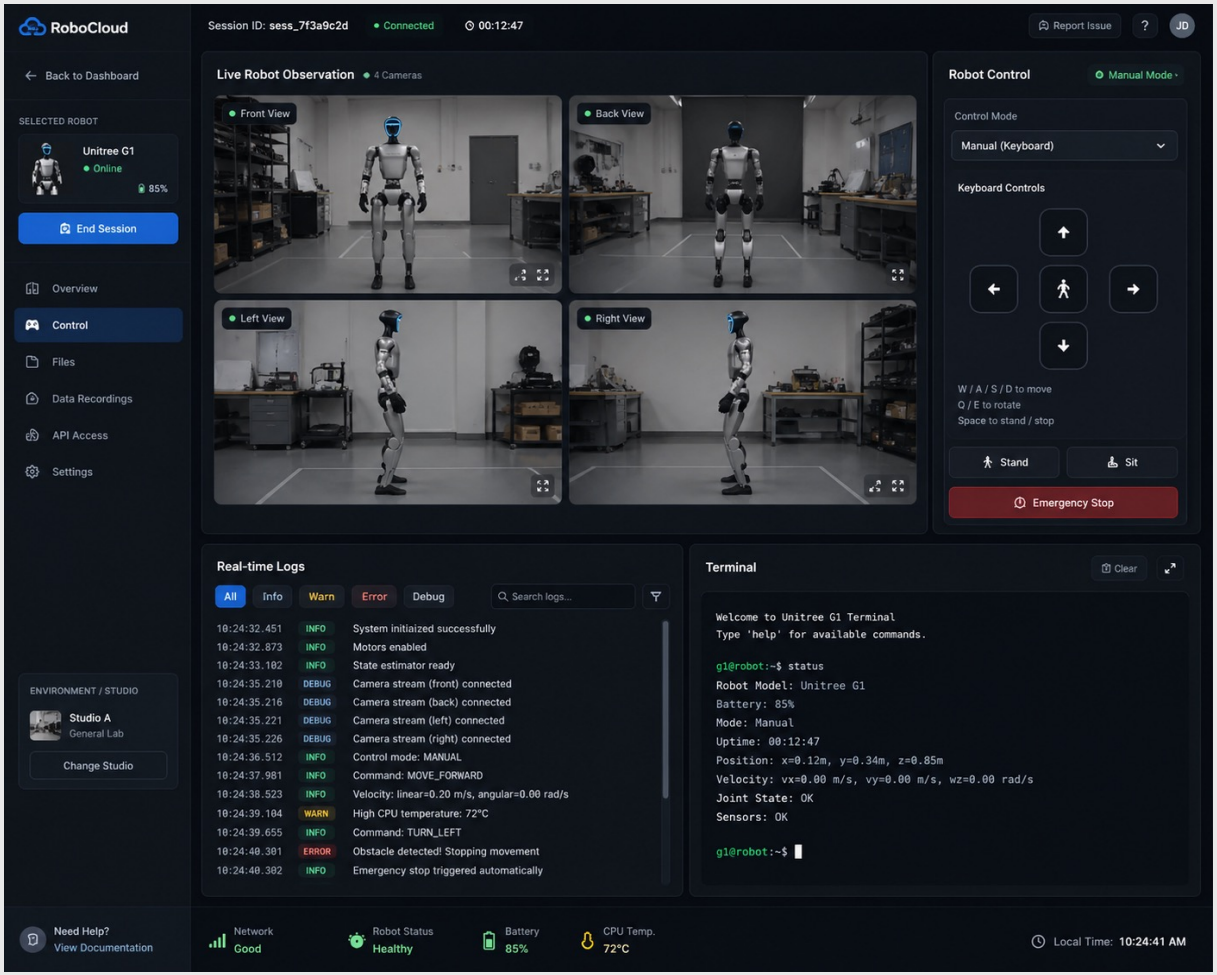
3. **Run Experiments Remotely**
   Access the robot in minutes via API or dashboard. Observe robot's behavior via studio's monitoring cameras and Control robots via APIs in real-time.

4. **Collect all experiments data**
   Collect all robots sensor/camera/LiDAR data and also studio monitoring videos for analysis and improvement.

KEY FEATURES

- **No hardware ownership**
  Access premium robots without capital expenditure. We handle maintenance, upgrades, and operations.
- **No setup required**
  Log in, configure your experiment, and start testing in minutes — not days. No travel. No hardware required.
- **Pay per use**
  Usage-based pricing by the hour. Scale up or down as your research needs change.
- **API-first access**
  Programmatic control and integration with existing ML pipelines and CI/CD workflows.

Product: Full-stack robotics infrastructure



Robot Fleet

- Multi-brand humanoid robots
- Standardized control interface

Studios

- Modular environments (stairs, doors, objects, lighting)
- Real-world testing scenarios
- Customization: Bring Your Own Piece

Software Platform

- Remote control & APIs
- Experiment recording & replay
- Data collection

Who use RoboCloud

Research Labs

& Universities

Academic & Independent Research

- Run experiments on-demand without owning hardware
- Reproducible physical benchmarks across robot platforms
- Ideal for PhD students, postdocs, and research teams

10×

faster iteration vs. lab robots

AI Engineers

& ML Teams

Embodied AI & Robotics Development

- Close the sim-to-real gap with real hardware testing
- API-first access integrates into existing ML pipelines
- Test vision, language, and control models physically

< 5 min

from login to first experiment

Enterprises

& Industrial R&D

Pre-Deployment Evaluation & Validation

- Validate robot platforms before committing to purchase
- Run real-world trials in controlled studio environments
- Scale large-scale tests and data collection without infrastructure

$0

capital expenditure to get started

From research to industrial validation.

A New Infrastructure Layer for Physical AI

MARKET SCALE

1 Billion+

humanoid robots by 2050

Every robot needs to be developed, tested, and validated — in the physical world. This is the Windows moment for embodied AI: a platform shift that will spawn thousands of new use cases.

Surging

investment in humanoid robotics

Capital flowing into robot platforms is accelerating. Hardware is maturing fast — but the infrastructure to test and deploy it at scale does not yet exist.

Critical

demand for real-world AI testing

As AI systems move from the digital to the physical world, simulation alone is no longer enough. Real-world validation becomes a hard requirement for every deployment.

EXPANSION OPPORTUNITIES

Proprietary Data Generation

Real-world robot interaction data is scarce and enormously valuable. Each session on RoboCloud generates training data that no simulation can replicate — building a proprietary dataset moat over time.

Cross-Platform Benchmarking

Standardized environments enable fair, reproducible performance comparisons across robot models. RoboCloud becomes the reference layer for evaluating embodied AI systems at scale.

Enterprise Validation-as-a-Service

Large companies evaluating robotics deployments need structured, repeatable trials before committing capital. RoboCloud provides the validation layer — reducing procurement risk across industries.

Every AI system interacting with the physical world needs this layer.

Usage-Based Revenue Model

REVENUE STREAMS

Pay-Per-Use

$40 – $100/hr

ENTRY POINT

varies by robot type & studio

FOR

Researchers, students, early-stage startups

Zero commitment. Log in, pick a robot, start testing in minutes. Billed by the hour — ideal for exploratory work.

Subscription Plans

Monthly/Annual

CORE VOLUME

prepaid hour bundles + priority access

FOR

AI teams, robotics startups, academia

Discounted hour bundles for recurring users. Includes priority scheduling, faster spin-up, and reserved studio slots.

Enterprise Contracts

Custom Pricing

HIGH VALUE

dedicated fleets, SLA, data licensing

FOR

Industrial R&D, large-scale validation

Dedicated robot allocation, custom environments, SLA guarantees, and proprietary data agreements for large-scale deployments.

KEY GROWTH DRIVER

UTILIZATION FLYWHEEL

Revenue scales with utilization

Robots are expensive assets that sit idle in labs most of the time. By aggregating demand across hundreds of users, RoboCloud runs robots continuously — turning capital assets into compounding infrastructure revenue.

UTILIZATION RATE

3–5×
higher utilization vs. owned robots

~$0
marginal cost per new user

∞
revenue upside with scale

AI Is Ready for the Physical World

FOUR CONVERGING FORCES

AI Breakthroughs

LLMs · Vision · Control

• READY

Foundation models have unlocked perception, reasoning, and dexterous control at levels unimaginable five years ago. The AI to power robots already exists.

Hardware Acceleration

More platforms · Lower costs

• ACCELERATING

A new generation of humanoid robot platforms is entering the market — from Figure to Unitree to Boston Dynamics. More hardware means more demand to test it.

Simulation Hitting Its Limits

Sim-to-real gap · Transfer failures

• CEILING REACHED

Simulation has matured — and so has the understanding of its ceiling. Teams now know that real-world testing is not optional; it is where products succeed or fail.

Investment Surge

$Billions flowing into embodied AI

• SURGING

Humanoid robotics received more venture and strategic capital in the last two years than in the prior decade combined. The ecosystem is forming fast — and needs infrastructure.

THE TIMING

CONVERGENCE POINT

The gap between AI capability and physical infrastructure is widest right now.

- 2017–23 — LLMs & vision models mature
- 2022–24 — Humanoid robot platforms emerge
- 2024 — Sim-to-real gap becomes critical
- NOW → **Infrastructure layer is missing**

RoboCloud is built for this exact moment.

Building the Robotics Infrastructure Layer

STRUCTURAL ADVANTAGES

01 Multi-Robot Platform Aggregation

Fleet diversity as a network effect

Supporting multiple robot brands in one platform creates a cross-platform standard no single manufacturer can offer. Each new robot added increases value for all users — and switching cost for any one of them.

• NETWORK EFFECTS

RELEVANCE

02 Proprietary Real-World Dataset

A data moat that compounds over time

Every experiment run on RoboCloud generates real interaction data — contact forces, sensor readings, failure modes — that simulation cannot replicate. This dataset becomes a durable, compounding asset no competitor can fast-follow.

• DATA ASSET

RELEVANCE

03 Developer Workflows & APIs

Deep pipeline integration

API-first access lets teams embed RoboCloud directly into their ML training loops, CI/CD pipelines, and evaluation suites. The deeper the integration, the higher the switching cost — and the more indispensable the platform becomes.

• INTEGRATION

RELEVANCE

COMPETITIVE POSITION

WHERE WE STAND

No one owns this layer yet.

- Robot manufacturers → hardware only, no access layer
- Simulation tools → digital only, no real-world bridge
- Cloud providers → no physical infrastructure expertise
- **RoboCloud → the only shared real-world access platform**

Manufacturing robot test at scale

VW scales its humanoid robots test with RoboCloud

CHALLENGE

- Volkswagen Group is exploring multiple industrial use cases for humanoid robots across manufacturing, logistics, and operations. Limited availability of physical robots creates bottlenecks—most facilities only have a few units shared across teams. Data scientists and robotics engineers lack consistent, on-demand access to hardware to validate AI models. Testing cycles are slow due to scheduling conflicts, setup time, and fragmented robot environments

SOLUTION

- VW leveraged RoboCloud to provide on-demand, remote access to a shared fleet of humanoid robots
- Standardized testing environments ("studios") were deployed to enable repeatable and scalable experiments across use cases
- Data scientists integrated their VLA/VLM models directly via RoboCloud APIs to run autonomous experiments without manual setup
- Built-in safety and control layers ensured secure execution of autonomous policies on physical robots
- Centralized monitoring, logging, and replay tools enabled rapid debugging and performance optimization

BENEFITS

✓ 60 percent reduction in operations costs
✓ Enabled parallel testing across multiple teams without increasing hardware investment
✓ Reduced experiment setup time from weeks to minutes
✓ Increased robot utilization significantly across R&D workflows
✓ Accelerated validation of AI models in real-world conditions

Volkswagen Group is one of the world's largest automotive companies, producing a wide range of cars, trucks, and mobility services through brands such as Volkswagen, Audi, Porsche, Škoda, SEAT, CUPRA, Bentley, Lamborghini, and MAN. It has a global footprint across Europe, the Americas, and Asia, with hundreds of thousands of employees worldwide.

"RoboCloud gives our teams immediate access to real robots, allowing us to scale experimentation across multiple use cases without being constrained by hardware availability."

Head of Robotics / AI, Volkswagen Group

Next Steps: Build the future of robotics together!

01 — 02 — 03

APPLY

Join the Private Beta Program

Request early access to RoboCloud and become one of our design partners.
We are onboarding a limited number of teams working on real-world robotics use cases.

DEFINE

Share Your Use Cases

Work directly with us to define:
your target applications
testing environments
robot capabilities needed
We co-design the platform based on your needs.

BUILD&TEST

Get Early Access & Run Experiments

Access the first version of RoboCloud to:
test your models on real robots
iterate faster with our support
shape the future of the platform